Business Problem Document

Customer Churn Analysis & Retention Strategy | Python + SQL + Power BI

1. Background

A telecom company sells broadband and mobile plans to about 7,200 customers across four regions of India. Roughly one customer in four has left. Acquiring a new customer costs far more than keeping an existing one, yet the retention team has no reliable view of why customers leave, who is likely to leave next, or how to spend a limited retention budget.

2. Business problem statement

Churn costs the company about a quarter of its monthly revenue, and retention effort is currently untargeted. The business needs (1) a clear explanation of what drives churn, (2) a model that ranks active customers by churn risk, and (3) a costed retention plan that says whom to contact and with which offer.

3. Objectives

- Clean the raw customer file and document every fix.
- Quantify churn rate and revenue lost; identify the strongest drivers with statistical tests.
- Build and compare three predictive models using cross-validation and a held-out test set.
- Choose the contact threshold that maximises expected campaign profit, under explicit cost assumptions.
- Produce a prioritised action list with a retention playbook per segment.
- Deliver a Python notebook, SQL query library and Power BI dashboard.

4. Key business questions

#	Question	Answered in
1	What is the churn rate and how much revenue does it cost?	SQL Q1, notebook s3, Power BI p1
2	Which customer attributes drive churn (contract, service, price, tenure, tickets, payment)?	Notebook s4, SQL Q2-Q8, Power BI p2
3	Who are the highest-risk active customers and how much revenue is at risk?	Notebook s9, SQL Q10-Q11, Power BI p3-p4
4	How accurate is the model and can it be trusted for decisions?	Notebook s6-8, SQL Q12, Power BI p5
5	Whom should we contact, with which offer, and what is the expected return?	Notebook s8-10, Power BI p4 simulator

5. Data

One row per customer (synthetic, generated with realistic churn drivers and deliberate quality problems). 7,260 raw rows reduce to 7,200 clean customers. Currency INR.

Group	Columns
Profile	customer_id, signup_date, gender, senior_citizen, partner, dependents, region, city
Account	tenure_months, contract (Month-to-month / One year / Two year), payment_method, paperless_billing
Services	phone_service, internet_service (Fiber optic / DSL / No internet), online_security, online_backup, device_protection, tech_support, streaming_tv, streaming_movies

Group	Columns
Billing & behaviour	monthly_charges, total_charges, avg_monthly_data_gb, support_tickets_6m, late_payments_12m
Target	churn (Yes = customer has left)

6. Data quality issues fixed

Issue	Rows	Fix
Duplicate customer_id	60	Keep first
Negative monthly_charges	40	Absolute value (sign error)
Missing total_charges	144	monthly_charges x tenure
Missing gender	108	Label Unknown
Inconsistent contract / payment text	~360	Trim and standardise

7. Approach

- **Clean and validate** (Python + SQL checks).
- **Explore**: churn rate by attribute with chi-square tests and Cramer's V for effect size.
- **Engineer features**: add-on count, charge per add-on, ticket intensity, new-customer flag.
- **Model**: logistic regression, random forest, gradient boosting; stratified 5-fold CV; untouched 25% test set; ROC-AUC and PR-AUC; calibration check.
- **Interpret**: odds ratios and permutation importance.
- **Decide**: threshold chosen by expected profit with explicit assumptions (INR 400 per contact, 30% offer success, 40% margin).
- **Act**: rule-based retention playbooks; revenue-at-risk list.
- **Report**: SQL queries and Power BI dashboard with a what-if campaign simulator.

8. Success criteria

- Model ROC-AUC clearly above 0.5 on unseen data, with probabilities that match actual churn rates.
- Targeted campaign has higher expected profit than contacting everyone.
- Every recommendation is tied to a number in the analysis.
- SQL and Python agree on shared figures.

9. Assumptions and limitations

- Data is synthetic; findings demonstrate method, not a real company.
- Churn is a snapshot label (left / stayed), so the model finds customers who *resemble* past churners; a time-to-event (survival) model would be the next step.
- Campaign economics (cost, success rate, margin) are assumptions and should be replaced with measured values from a pilot.
- Association is not causation: the model shows who is at risk, not proof that changing a feature will keep them.